

Effect of Multi-Dimensional Zinc Oxide on Electrical Properties of Polypropylene Nanocomposites for HVDC Cables

Jun-Wei Zha,^{1,2} Qing-Qing Qin¹ and Zhi-Min Dang²

¹School of Chemistry and Biological Engineering, University of Science and Technology Beijing
Beijing 100083, P. R. China

²State key Laboratory of Power System, Department of Electrical Engineering,
Tsinghua University, Beijing 100084, P. R. China

ABSTRACT

Polypropylene (PP) is one of the most important high voltage direct current (HVDC) cable insulation materials, which has been widely used in recent years. However, the accumulation of space charge is the most notable problem to limit the practical application of PP. In this paper, the PP nanocomposites filled with polystyrene-*b*-poly(ethylene-co-butylene)-*b*-polystyrene (SEBS) and multi-dimensional zinc oxide composite fillers (co-ZnO) prepared by melt blending method showed excellent properties due to the good compatibility between the co-ZnO filler and PP/SEBS blends. The dielectric permittivity of the composites can be up to 3.1 with 1 wt% doping of co-ZnO. Compared with PP/SEBS, the introduction of co-ZnO filler can largely suppress the space charge accumulation as well as the electric field distortion, and improve the breakdown strength by 14%. It is promising to make the PP/SEBS/co-ZnO composites as one better candidate for HVDC cable insulation materials.

Index Terms — HVDC cables, space charge, PP composites, multi-dimensional

1 INTRODUCTION

HIGH voltage cables have become the main transmission medium for the development of transmission systems [1-3]. The high voltage direct current (HVDC) transmission system has the characteristics of convenient connection, long distance, high voltage and low loss, which has been widely used for power transmission.

There are two types of cable materials used in HVDC; namely, impregnated and extruded polymer insulated cables. However, the manufacturing process of the first type of cable materials is complicated. The operation and maintenance cost is high and the cables are easy to be damaged in use [4,5]. In contrast, the polymer insulated cable is an ideal cable material which is suitable for HVDC transmission cables. The internal insulating medium is mainly polyolefin. It has the advantage of light weight, flexibility, easy installation and low cost. It also can be prepared by a simple method. Cross-linked polyethylene (XLPE) as one of the conventional insulating materials always exists some disadvantages. The equipment used for cross-linking process will occupy a large space, low production efficiency, large energy consumption, equipment investment and energy loss, especially hard to be recycled. In recent years, the polypropylene (PP) has attracted more

attention because of its high melting point, excellent insulation properties and strong corrosion resistance. In particular, the PP can be recycled and reused after using to avoid pollution, which not only saves the resources, but also is environmentally friendly. Therefore, PP has become one of the most promising materials for large-capacity HVDC cables.

The accumulation of space charge in polymer insulation materials has become a big issue to limit its lifetime and ranges of use [6,7]. The electrode injection is generally considered to be the major factor in the formation of space charge. Polymer has lots of traps distributed within the material because of its inherent structural characteristics, which can capture the charges injected by the electrodes [8,9]. When the number of accumulated charge is sufficient, it may occur a partial discharge in the material, which can cause the polymer chains to break or even degrade, generate low molecular and gaseous molecules. Then, it will provide cracks and new traps for charge accumulation until the material structure is severely damaged distortion [10,11]. This phenomenon is also called aging which can affect the conductivity and breakdown of the insulating material, thus affecting the lifetime of cable materials.

At present, there are three methods proposed to suppress the space charge in extruded DC cable insulation materials: blending, grafting and adding nanoparticles. In particular, a small amount of nanoparticles can significantly improve

properties of polymer materials due to its surface effect, quantum size effect and small size effect [12,13]. When the particle size is reduced to nanoscale, the effect of interface will become important. Many of the excellent properties of nanodielectrics are considered to be related to interface structure and behavior. By doping different kinds of nanoparticles, the electrical, mechanical and other properties of the material will be changed at some extent. The recent researches reported that the fillers are mainly use ZnO, MgO and Al_2O_3 nanoparticles and so on. Besides, ZnO nanoparticles can also be used as UV inhibitors effectively improve the service life of polymer insulation materials [14-16]. Some studies reported that when one-dimensional ZnO nanowires are doped into an insulating material, the energy level barrier formed by the nanowires and the polymer matrix is stronger than the nanoparticles and the polymer matrix done respectively, which makes it more difficult for carriers to migrate [17,18]. Therefore, it can well suppress the space charge accumulation, which will be widely used in the preparation of composite insulation materials.

In this paper, the ZnO composite filler (co-ZnO) was synthesised by solvothermal method in order to avoid agglomeration of fillers due to excessive doping. The PP nanocomposites filled with thermoplastic elastomer and co-ZnO were prepared by melt blending method. The composite filler on space charge distribution, as well as volume resistivity, dielectric properties and other performances of PP composites are discussed.

2 EXPERIMENT PROCEDURES

2.1 RAW MATERIALS

The block PP copolymer (K1008) with melt index (MI) of 8g/10min was supplied by Sinopec Beijing Yanshan Company. Thermoplastic elastomer, namely polystyrene-*b*-poly(ethylene-co-butylene)-*b*-polystyrene (SEBS, 6150) was purchased from Taiwan Yu Industrial Co., Ltd. The spherical ZnO particles with average diameter of 50 nm and ZnO nanowires with average diameter of 80 nm and length of 600-800 nm were obtained from Beijing Dk Nano technology Co., Ltd.

2.2 PREPARATION OF CO-ZNO FILLERS

The co-ZnO composite fillers were synthesised through solvothermal method. Firstly, the NaOH ethanol solution (0.5 mol/L) were prepared. The 0.0015 mol ZnO nanoparticles and 0.0015 mol ZnO nanowires were slowly added to the NaOH ethanol solution above. The obtained mixture was treated by ultrasonic dispersion for 1 h. Then, the mixture was transferred into a reactor, which was sealed and maintained at 170 °C for 12 h. After cooling to room temperature and the centrifuging process, the product was washed with ethanol and deionized water for several times to absolutely remove the residual NaOH solution. Finally, the washed products were treated by vacuum drying at 60 °C for 8 h. Finally, the co-ZnO composite fillers were obtained.

The prepared co-ZnO fillers were dispersed into the PP and SEBS using an torque rheometer (RM-200C, HAPRO) at 190

°C for 15 min under rotor speed of 60 rpm. The mass contents of co-ZnO fillers added were set as 0, 0.5, 1.0 and 1.5 wt%. The PP/SEBS/co-ZnO composites were prepared through hot pressing method at 190 °C under 15 MPa for 15min and slowly cooled to room temperature.

2.3 CHARACTERIZATION

The morphologies of ZnO nanoparticles, ZnO nanowires, co-ZnO and their dispersion in the PP and SEBS were observed using filed emission scanning electron microscopy (FESEM, Hitachi SU8010).

The melting behaviors of the PP/SEBS and their composites were characterized through differential scanning calorimetry (DSC) at heating rate of 10 °C /min under gas N_2 after one heating-cooling cycling process to eliminate thermal history of the samples.

Space charge behaviour of the film samples was examined using pulsed electro-acoustic (PEA) method under applied electric field of 60 kV/mm for 90 min at room temperature. Before testing, all the samples were dried in an oven at 70 to remove the humidity. A pulse voltage with frequency of 460 Hz and amplitude of 5 ns was applied to the samples with the average thickness of 250 μm .

Dielectric properties of the samples were measured by a high-resolution ALPHA analyzer (Novocontrol, Germany). The samples with average thickness of 1 mm were plated with copper electrodes on their upper and lower sides. The dielectric properties were tested in the frequency ranging from 10^2 to 10^6 Hz at 25 °C.

DC breakdown strength was tested by dielectric instrument using the sphere-sphere electrode configuration at room temperature. The samples with average thickness of 100 μm were immersed in silicone oil to isolate from the air. Twelve samples were tested for each kind of sample to calculate the characteristic strength by Weibull distribution.

DC conductivities of the samples were measured under applied DC voltage of 1 kV via a three-electrode system at room temperature. The current was recorded by using an electrometer (KEITHLEY 6517B).

Mechanical properties of the samples were tested by an electronic universal testing system (Instron 5900) under tensile rate of 250 mm/min.

3 RESULTS AND DISCUSSIONS

3.1 MICROSTRUCTURAL CHARACTERIZATION OF THE PP/SEBS/CO-ZNO COMPOSITES

SEM images of the nanoparticles, nanowires and composite fillers are shown in Figure 1a~1d. It can be seen that the ZnO nanoparticles with diameter of 50 nm are evenly dispersed as shown in Figure 1a. The ZnO nanowires with ~600 nm in length as shown in Figure 1b. The co-ZnO composite fillers are well synthesised as shown in Figure 1c and 1d. Figure 1c illustrates the uniform attachment of nanoparticles on a linearized structure (Figure 1d is the magnification image from Figure 1c). Figure 1e shows the SEM images of fracture

surface of PP composites. It can be seen that the co-ZnO fillers have a good uniform distribution in the PP/SEBS matrix.

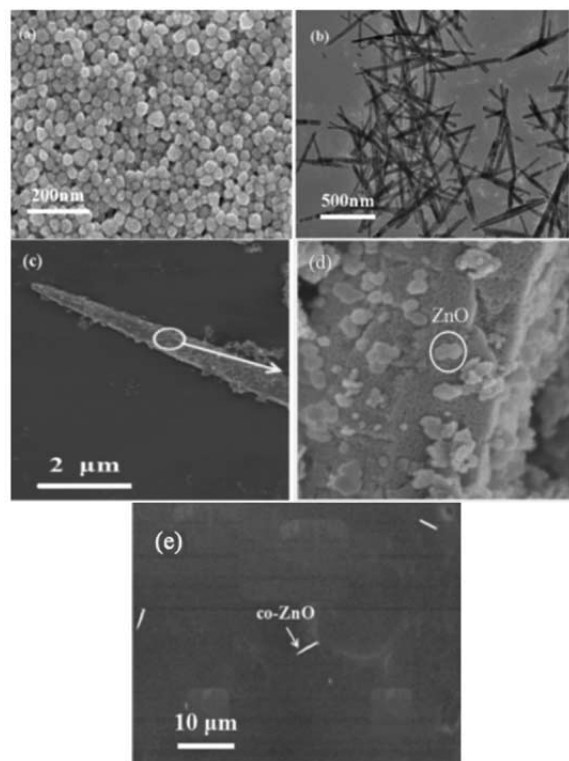


Figure 1. SEM images of (a) ZnO nanoparticles, (b) ZnO nanowires, (c) and (d) co-ZnO composite fillers, (e) PP composites with the co-ZnO loading.

3.2 DSC CURVES OF PP/SEBS/CO-ZNO COMPOSITES

Figure 2 shows the melt behavior of PP/SEBS blends and their composites with different concentration of co-ZnO fillers. Under nitrogen atmosphere, the samples were heated from 30 to 200 °C at a rate of 10 °C/min and remained at 200 °C for 2 min followed by cooling process. Table 2 lists the melting points of the materials corresponding to their melting peaks in Figure 2. It can be found that the melting points of the PP/SEBS blends and their composites are almost close to 163 °C. That is, there is no obvious variation of the melting points with the addition of co-ZnO, which is attributed to good compatibility of the co-ZnO filler with the matrix. However, their melting heat (ΔH_f) calculated from the DSC curves initially increase with increasing the co-ZnO concentration to 1.0 wt%, and then decrease with 1.5 wt% co-ZnO loading. Generally, the crystallinity (X_c) can be estimated from an empirical equation as follows:

$$X_c = \frac{\Delta H_f}{\Delta H_c} \times 100\% \quad (1)$$

where ΔH_c is defined as the melting heat when the degree of crystallization of PP is 100% and its value is set as 165 J/g [19].

It can be seen from Table 1 that the ΔH_f and X_c of the composites with 1.0 wt% co-ZnO filled is 69.1 and 41.8%,

respectively, which is larger than that of PP/SEBS blends. The addition of co-ZnO as nucleating agent affects the regularity of molecular or the crystallization of the materials.

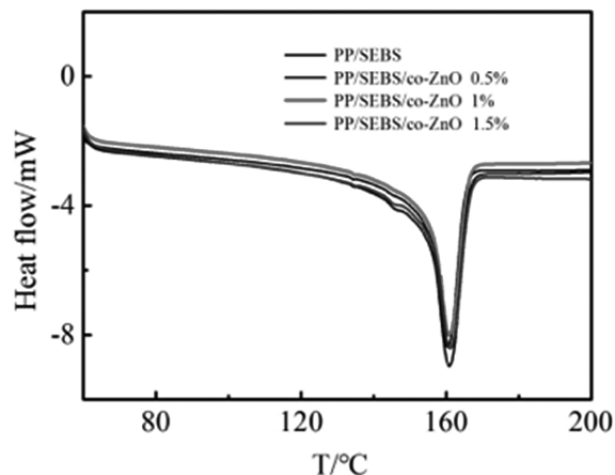


Figure 2. DSC spectra of the PP/SEBS/co-ZnO composites.

Table 1. Melting point and melting heat for the composites.

co-ZnO content (wt%)	T_g	ΔH_f (J/g)	X_c (%)
0	163.54	67.12	40.6
0.5	163.68	68.42	41.5
1	163.55	69.1	41.8
1.5	163.32	67.15	40.6

3.3 DIELECTRIC PROPERTIES OF PP/SEBS/CO-ZNO COMPOSITES

Figure 3 shows the frequency dependence of dielectric permittivity and dielectric loss of the PP/SEBS blends and their composites with different concentration of co-ZnO fillers. It is found that the addition of co-ZnO fillers can give rise to relatively larger permittivity, compared to the PP/SEBS blends. The dielectric permittivity of the composites reach to 3.1 at 1 wt% co-ZnO fillers doping, which is 8.8% higher than that of the PP/SEBS blends. It is noteworthy that there is significant decrement in the dielectric loss of the composites with the co-ZnO filled as shown in Figure 3b. The dielectric loss decreases to about 5×10^{-3} in the frequency ranging from 10^2 to 10^6 Hz, which is much lower than that of pure PP/SEBS blends (>0.01).

3.4 SPACE CHARGE OF THE PP/SEBS/CO-ZNO COMPOSITES

Generally, the space charge is formed mainly including electrode injection and impurity ionization. When the applied DC electric field exceeds the critical injection electric field of the dielectric, the electrode would inject the space charge with the same polarity into the medium. Impurity ionization is the ionization form that the impurities, by-products, additives and other substances ionized under an applied electric field. These impurities are introduced during the process of the raw material production and vulcanization molding [20-22]. The

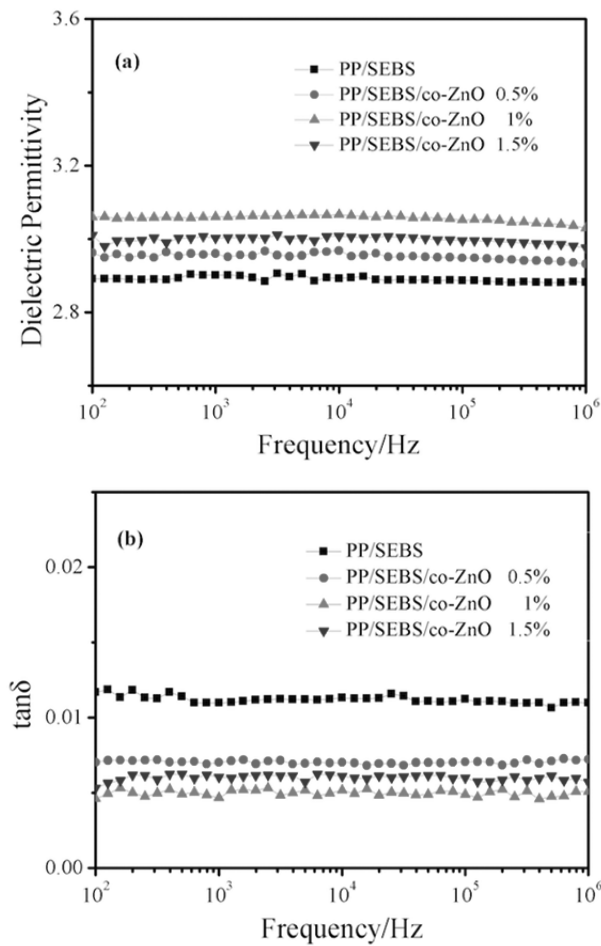


Figure 3. Frequency dependence of (a) dielectric permittivity and (b) dielectric loss tangent of PP/SEBS/co-ZnO composites.

space charge accumulation of PP composites depends on the trapping, releasing and transporting properties of the charge carriers inside the material. The charge trap depth, trap density and charge carrier mobility and other factors will cause changes in the space charge accumulation [23-25]. More studies have found that a large number of charge trapping sites can be introduced on the interface between the nanoparticles and the matrix by doping the nanoparticles in the composite. This is mainly due to the charge transport behavior, which has a great effect on the space charge accumulation in the composite.

Figure 4 shows space charge behavior of the composites under DC electrical field of 60 kV/mm. It can be seen from Figure 4a that there is a large amount of space charge injected near the cathode and anode after adding elastomer SEBS in the PP. With the increasing volts-on time, more charges would be injected near the cathode. The amount of space charge accumulation increases with the increase duration time. However, when the co-ZnO is added into the PP/SEBS blends, the space charge accumulation in the PP composites is suppressed. For the PP/SEBS/co-ZnO composites with 1 wt% co-ZnO loading, it is found that only a very small amount of electrode charge and ionization of the heterogeneous charge are formed as shown in Figure 4c. This phenomenon may be

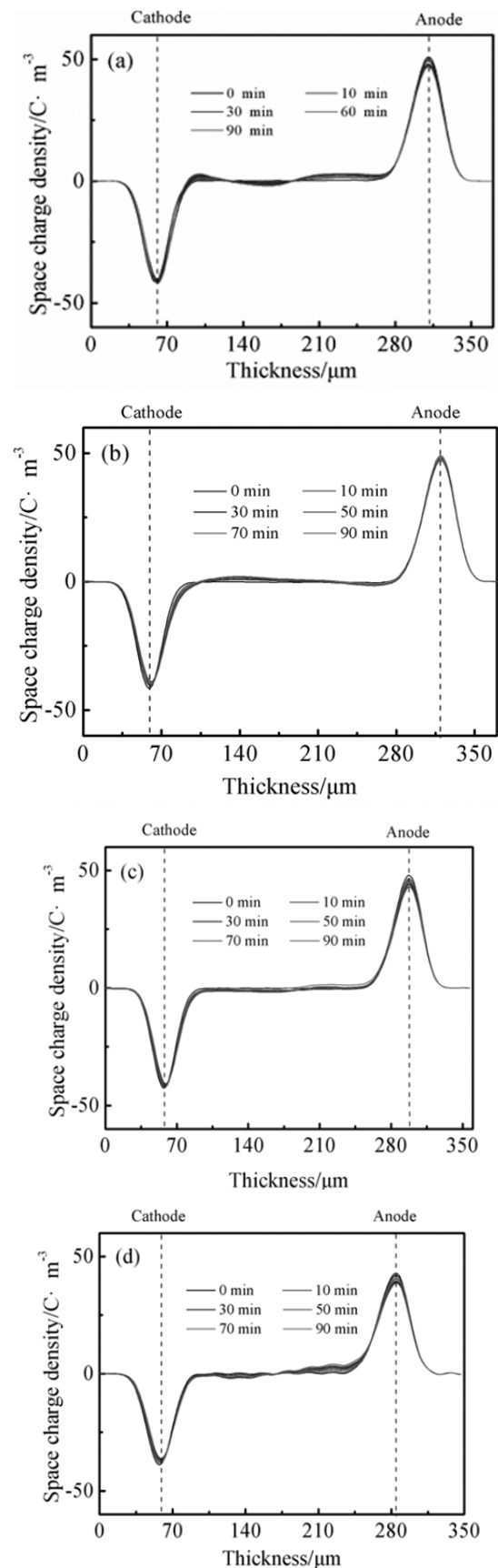


Figure 4. Space charge distribution in the PP composites under the applied electric field of 60 kV/mm with (a) 0, (b) 0.5 wt%, (c) 1.0 wt% and (d) 1.5 wt% co-ZnO loading.

attributed to the introduction of co-ZnO fillers which cause a certain number of traps at the interfaces between the fillers and matrix, as well as the interfaces between the ZnO nanofibers and nanoparticles. The deep traps can capture charge carriers, which resulted in the formation of local electrical fields, thus increasing the potential barrier for charge carrier injection to show the excellent space charge suppression effect. Significant suppression of space charge of the PP composites is of great importance to the development of HVDC cables. Therefore, the doping of co-ZnO fillers into polymeric insulating materials has been manifested to mitigate space charge accumulation with the synergistic effect between the ZnO nanofibers and nanoparticles and polymer matrix which can effectively suppress the motion of carriers as well as the recombination.

Figure 5 shows the electric field distribution caused by space charges in PP/SEBS/co-ZnO composites. As shown in Figure 5a, the electric field of PP composite without co-ZnO fillers is seriously deviated by about 5.5 kV/mm due to the accumulation of space charge inside the materials. However, the electric field deviation is decreased with the addition of co-ZnO. The lowest deviation of PP/SEBS/co-ZnO composites with 1 wt% co-ZnO loading is 1.1 kV/mm as shown in Figure 5c. The low electric field strength deviation helps to suppress the generation and growth of space charge in high-voltage cables during long-term use, thus avoiding the breakdown of insulating materials, protecting the overall performance of materials and prolonging the service life.

3.5 MECHANICAL PROPERTY OF PP/SEBS/CO-ZNO COMPOSITES

Figure 6a shows the stress-strain curves of PP/SEBS/co-ZnO composites with different content of co-ZnO. Figure 6b is the corresponding histogram of tensile strength and elongation at break values. With the increase of co-ZnO loading, the tensile strength and elongation at break both increase and then decrease.

When the content of co-ZnO is 1 wt%, the PP/SEBS/co-ZnO composite has an elongation at break of 606.7% and a tensile strength of 32.3 MPa, which has a small effect on the overall mechanical properties of the composites. This phenomenon is attributed to the huge nanoparticle-polymer interface in the composite, thereby ensuring good compatibility and interfacial adhesion between the fillers and the polymer matrix, resulting in improved mechanical properties. It is noteworthy that the mechanical properties of the composites decrease slightly when the filler content reaches 1.5 wt%. The deterioration of mechanical properties is attributed to the aggregation of co-fillers, which easily leads to the negative effect of defects and stress concentration. And the dispersion of fillers in the matrix also has great influence on the mechanical properties of the composites. The PP/SEBS/co-ZnO with 1 wt% co-ZnO loading has the best mechanical properties.

3.6 BREAKDOWN STRENGTH OF PP/SEBS/CO-ZNO COMPOSITES

DC breakdown strength is an important electrical parameter for dielectric materials. Relevant previous studies have shown

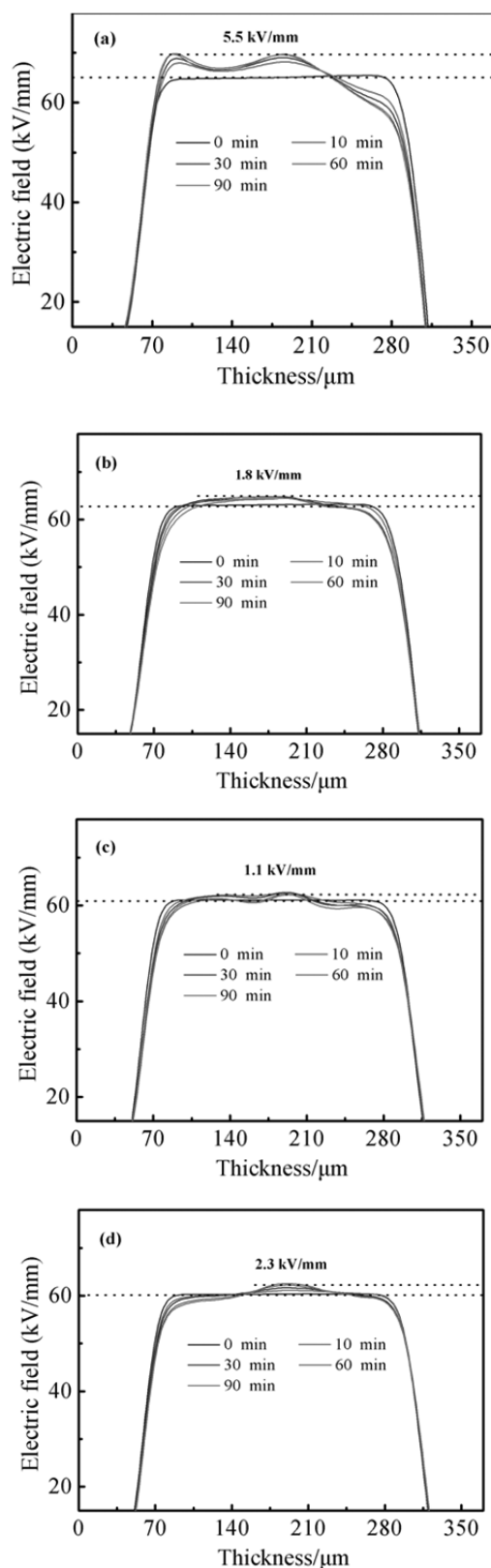


Figure 5. Electric field distribution in PP/SEBS/co-ZnO composites under 60 kV/mm electric field with co-ZnO loading (a) 0, (b) 0.5 wt%, (c) 1.0 wt% and (d) 1.5 wt%.

that when high-voltage electric field was applied to the materials, the charges were injected into the interior of the materials and the impurities were ionized. Subsequently, the

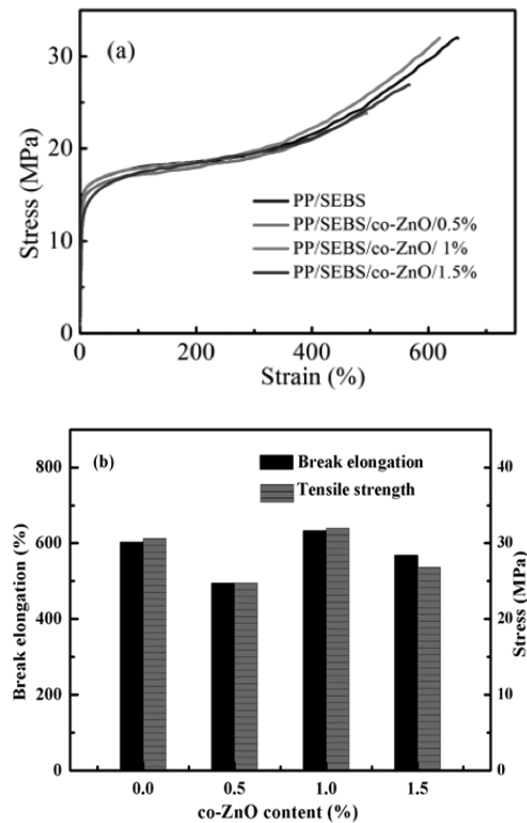


Figure 6. Tensile stress-strain of PP/SEBS/co-ZnO composites (a) curves and (b) histogram.

energy of the charge carriers was accelerated in the DC electric field. When the energy is accelerated sufficiently high, it will hit or even destroy the molecular chains of the polymer, causing a partial discharge which eventually leads to the breakdown of the insulating materials. However, when the fillers are doped into the polymer matrix, on one hand, the deep traps can be formed at the interface between the polymer matrix and the fillers to capture the charges more effectively. On the other hand, the fillers also act as scattering sources to reduce the charge carrier energy, thus preventing the generation and growth of electrical tree.

Figure 7 shows the Weibull plots of DC breakdown strengths of PP/SEBS blends and PP/SEBS/co-ZnO composites with different contents of co-ZnO. All of the testing samples have a film thickness of about 100 μm . As shown in Figure 7, it can be seen that a certain amount of co-ZnO can improve the DC breakdown strength to a certain extent compared with the polymer matrix. With the increase of the co-ZnO loading, the breakdown strength of the composite first increases and then decreases, and the composite with 1 wt% co-ZnO loading has the best breakdown strength of 240 kV/mm. When the content of co-ZnO is 1.5 wt%, the breakdown strength shows a decrease to about 180 kV/mm, which can be explained that with the increase of composite filler loading, there may be many large aggregates. It can be seen from the space charge and the electric field deviation curves, in which the space charge accumulation of the composite with 1.5 wt% co-ZnO loading is more serious, and

the corresponding electric field deviation is more obvious, which also resulting in reducing the electric breakdown strength of the composites.

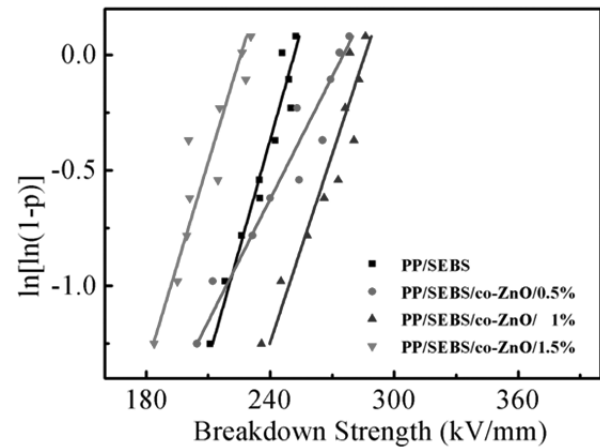


Figure 7. Weibull distribution plot of breakdown strength of the PP/SEBS/co-ZnO composites.

3.7 CONDUCTIVITY CHARACTERISTICS OF PP/SEBS/CO-ZNO COMPOSITES

Figure 8 shows the current, current density and DC conductivity of the PP/SEBS/co-ZnO composite with co-ZnO loading. Figure 8a and 8b show the current and current density curves of PP/SEBS/co-ZnO composites versus time. It can be seen that the current and current density of the PP/SEBS/co-ZnO composites decrease when a certain amount of filler is doped. It is due to the co-ZnO fillers being polarized at the interfaces and producing electrical potential wells under DC electric fields, which trap the traps and reduce the transport of charge carriers within the material. Figures 8c and 8d show that the conductivity decreases to a minimum of 1.1×10^{-16} S/m when the co-ZnO content is 1 wt%. However, with the increasing co-ZnO, the agglomeration is obvious and the adhesion with the matrix will decrease. The carriers are easily transported by electron tunneling effect with the formation of free conductive path, thus making the conductivity increases.

4 CONCLUSIONS

In this paper, the composite filler co-ZnO was successfully synthesized by solvothermal method. The ZnO nanoparticles with a diameter of 50 nm were successfully attached to the ZnO nanowires, both of which formed a core-shell structure. The PP/SEBS/co-ZnO composites were successfully prepared. The conclusions are as follows:

- (1) The PP/SEBS/co-ZnO composites with 1 wt% co-ZnO loading can effectively inhibit the space charge accumulation, which may be attributed to the introduction of deep traps. The increasing of trap density and depth strengthen the ability to capture space charge and meanwhile increase the potential well. Thus, the composites exhibit excellent space charge suppression.

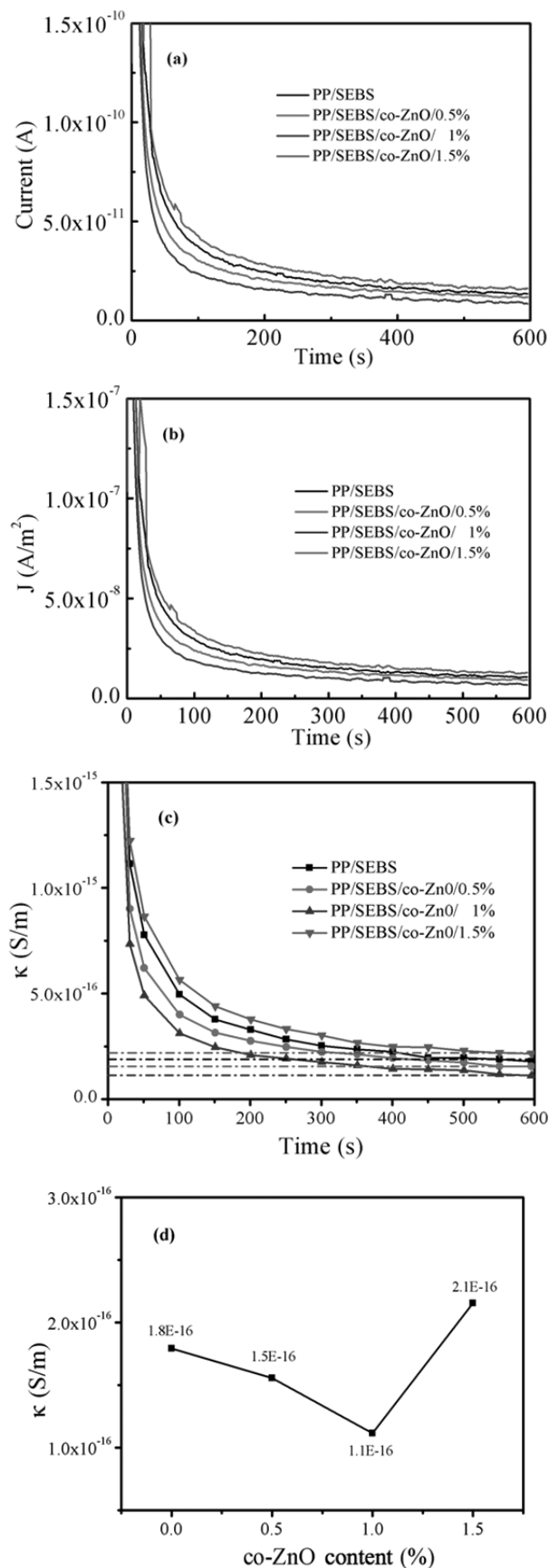


Figure 8. The (a) current curves and (b) current density curves of PP/SEBS/co-ZnO composites, and conductivity curves of PP/SEBS/co-ZnO composites (c) versus time and (d) versus co-ZnO content.

(2) The PP/SEBS/co-ZnO composites show stable dielectric permittivity and relatively low dielectric loss in the frequency range of $10^2 \sim 10^6$ Hz. The PP/SEBS/co-ZnO composites with 1 wt% co-ZnO loading also show the highest breakdown strength of 240 kV/mm.

(3) The PP/SEBS/co-ZnO composites with 1 wt% co-ZnO loading show the lowest current and conductivity.

In summary, the PP/SEBS/co-ZnO composites as environmentally friendly insulation have good application prospects, especially in the renewable high voltage DC transmission line.

ACKNOWLEDGMENT

The authors gratefully acknowledge support from the National Natural Science Foundation of China (Nos. 51622701 and 51425201), National Basic Research Program of China (973 Program, No. 2014CB239501), Beijing Nova program (Z181100006218006), the Fundamental Research Funds for the Central Universities (No. FRF-TP-16-001C1).

REFERENCES

- [1] R. J. Fleming, "Conductivity and space charge in LDPE/BaSrTiO₃ nanocomposites," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 18, pp. 15–23, 2011.
- [2] T. Yamanaka, S. Maruyama, and T. Tanaka, "The development of DC+/-500 kV XLPE cable in consideration of the space charge accumulation," *IEEE Int. Conf. Prop. Appl. Dielectr. Mat. (ICPAM)*, 2003, vol. 2, pp. 689–694, 2003.
- [3] G. Mazzanti, M. Marzotto, *Extruded cables for high-voltage direct-current transmission: advances in research and development*, Wiley & Sons, pp. 88–92, 2013.
- [4] M. Praeger, Z. Xu, P. Lewin, G. Chen, "Partial discharge in mass-impregnated, lapped PPLP cables," *IEEE Electr. Insul. Conf.(EIC)*, 2018, pp. 403–406.
- [5] G. Chen, M. Hao, Z. Xu, A. Vaughan, J. Cao, H. Wang, "Review of high voltage direct current cables," *CSEE J. Power Energy Sys.*, vol. 1, pp. 9–21, 2015.
- [6] T. J. Lewis, J. P. Llewellyn, "Electrical conduction in polyethylene: The role of positive charge and the formation of positive packets," *J. Appl. Phys.*, vol. 113, no. 22, pp. 223705, 2013.
- [7] J. W. Zha, Z. M. Dang, H. T. Song, Y. Yin, G. Chen, "Dielectric properties and effect of electrical aging on space charge accumulation in polyimide/TiO₂ nanocomposite films," *J. Appl. Phys.*, vol. 108, pp. 094113, 2010.
- [8] C. Liang, P. Song, H. Gu, et al., "Nanopolydopamine Coupled Fluorescent Nanozinc Oxide Reinforced Epoxy Nanocomposites," *Compos. A: Appl. Sci. Manufact.*, vol. 102, 126–136, 2017.
- [9] J. W. Zha, Y. H. Wu, S. J. Wang, D. H. Wu, H. D. Yan and Z. M. Dang, "Improvement of Space Charge Suppression of Polypropylene for Potential Application in HVDC Cables," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 23, pp. 2337–2343, 2016.
- [10] S. Kumara, B. Ma, Y. V. Serdyuk, S. M. Gubanski, "Surface Charge Decay on HTV Silicone Rubber: Effect of Material Treatment by Corona Discharges," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 19, no. 6, pp. 2189–2195, 2012.
- [11] D. Pitsa, M. G. Danikas, G. E. Vardakis and T. Tanaka, "Influence of homocharges and nanoparticles in electrical tree propagation under DC voltage application," *Electr. Eng.*, vol. 94, pp. 81–88, 2012.
- [12] G. Mazzanti, G. Chen, J. C. Fothergill, N. Hozumi, J. Li, M. Marzotto, F. Mauseth, P. Morshuis, C. Reed, A. Tzimas and K. Wu, "A Protocol for Space Charge Measurements in Full-size HVDC Extruded Cables," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 22, pp. 21–34, 2015.
- [13] S. Li, D. Min, W. Wang, G. Chen, "Linking traps to dielectric breakdown through charge dynamics for polymer nanocomposites," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 23, pp. 2777–2785, 2016.

- [14] S. Luo, S. Wang, Y. Tu, C. Wang and S. Qin, "Direct current electrical characteristics of nano-CB/LDPE composite," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 24, pp. 1390–1395, 2017.
- [15] N. M. K. Abdel-Gawad, A. Z. E. Dein, D. A. Mansour, H. M. Ahmed, M. M. F. Darwish and M. Lehtonen, "Enhancement of dielectric and mechanical properties of Polyvinyl Chloride nanocomposites using functionalized TiO₂ nanoparticles," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 24, pp. 3490–3499, 2017.
- [16] C. Liang, P. Song, H. Gu, C. Ma, H. Zhang, X. Xu, Q. Zhang and J. Gu, "Nanopolydopamine Coupled Fluorescent Nanozinc Oxide Reinforced Epoxy Nanocomposites," *Compos. A: Appl. Sci. Manufact.*, vol. 102, pp. 126–136, 2017.
- [17] C. D. Green, A. S. Vaughan, G. C. Stevens, A. Pye, S. J. Sutton, T. Geussens and M. J. Fairhurst, "Thermoplastic Cable Insulation Comprising a Blend of Isotactic Polypropylene and a Propylene-ethylene Copolymer," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 22, pp. 639648, 2015.
- [18] Y. Yin, J. Chen, J. Yang, and D. Xiao, "Effect of space charge in nanocomposite of LDPE/TiO₂," *IEEE Int. Conf. Prop. Appl. Dielectr. Mat. (ICPADM)*, 2003, vol. 3, pp. 913–916.
- [19] J. W. Zha, Y. Wang, W. K. Li, S. J. Wang, and Z. M. Dang, "Electrical Properties of Polypropylene/Styrene-Ethylene-Butylene-Styrene Block Copolymer/MgO nanocomposites," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 24, pp. 1457–1464, 2017.
- [20] N. Gonzalez, J. R. Riba, M. D. A. Custal, E. Armelin, "Improvement of insulation effectiveness of natural rubber by adding hydroxyl-functionalized barium titanate nanoparticles," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 24, pp. 2881–2889, 2018.
- [21] J. W. Zha, H. T. Song, Z. M. Dang, C. Y. Shi, and J. Bai, "Mechanism analysis of improved corona-resistant characteristic in polyimide/TiO₂ nanohybrid films," *Appl. Phys. Lett.*, vol. 93, pp. 192911-1-3, 2008.
- [22] J. W. Zha, Q. Cheng, H. D. Yan, W. K. Li, Z. M. Dang, "Effect of multi-structured zinc oxide on the electrical properties of polypropylene insulating materials," *J. Phys. D: Appl. Phys.*, vol. 50, pp. 305301-1-5, 2017.
- [23] J. W. Zha, H. D. Yan, W. K. Li, and Z. M. Dang, "Morphology and crystalline-phase-dependent electrical insulating properties in tailored polypropylene for HVDC cables," *Appl. Phys. Lett.*, vol. 109, pp. 222902, 2016.
- [24] J. Hao, Y. Li, R. Liao, G. Liu, Q. Liao, C. Tang, "Fabrication of Al₂O₃ Nano-Structure Functional Film on a Cellulose Insulation Polymer Surface and Its Space Charge Suppression Effect," *Polymers*, vol. 9, pp. 502-1-19, 2017.
- [25] Q. G. Chi *et al.*, "Nano iron oxide-deposited calcium copper titanate/polyimide hybrid films induced by an external magnetic field: toward a high dielectric constant and suppressed loss," *J. Mater. Chem. C*, vol. 4, pp. 8179–8188, 2016.



J. W. Zha (M'17) received Ph. D. degree in materials science and engineering in 2010 from Beijing University of Chemical Technology. He studied in University of Southampton, as a Joint-Ph.D student sponsored by China Scholarship Council from 2009 to 2010. He currently works in the Department of Polymer Science and Engineering, University of Science & Technology Beijing. His research interests focus on nanodielectrics and polymer nanocomposites for electrical engineering. He has published more than 80 journal papers.



Q. Qin received her Bachelor degree in chemistry from Xuchang University in 2016. She currently studies in the Department of Polymer Science and Engineering, University of Science & Technology Beijing for Master degree.



Z. M. Dang received his Ph. D. degree in electrical engineering from Xi'an Jiaotong University in 2001. He is currently a Professor in Department of Electrical Engineering, Tsinghua University, China. He is also a member of IEEE on Dielectric and Electrical Insulation. His present research interests are Advanced Energy/Electrical Materials and Systems. He has published more than 200 journal papers with a current citation record of over 5000 times and an H-index of 41.